PRE-ALGEBRA (8TH) · UNIT 7: GEOMETRY

Skate Park Design

Use the Pythagorean Theorem, distance, and volume to design a skate park.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The town is building a new skate park, and the design team wants student input. Ramps must have the right lengths, paths must be planned on the map, and the concrete must be ordered in the right amount.

You will use right triangles, the distance formula, and volume formulas to create a blueprint and a concrete order.



Ramp
rises 5 ft over 12 ft

Bowl
cylinder: radius 10 ft
depth 6 ft

Planters
cones: radius 3 ft, height 4 ft

Learning Goals

- I can use the Pythagorean Theorem and its converse in real situations.
- I can use the Pythagorean Theorem to find distances on the coordinate plane.
- I can use volume formulas to plan materials.

Project Timeline

When	What you will do
Day 1	Building Ramps
Day 2	Planning Paths
Day 3	Ordering Concrete
Final	Skate Park Blueprint + reflection

Part 1: Building Ramps

Day 1 · Use the Pythagorean Theorem.

1. A 10-foot ramp covers a base of 8 feet. How high does it rise?

2. A 15-foot rail reaches 9 feet high. How far along the ground does it reach?

3. A builder measures a frame with sides 9, 12, and 16 feet. Is the corner a right angle?

Part 2: Planning Paths

Day 2 · Each grid unit on the map is 5 feet.

4. Find the distance between (−3, 4) and (5, −2) in units and in feet.

5. Is the triangle with vertices (0, 0), (6, 0), and (6, 8) a right triangle? How long is the longest side?

6. How much shorter is the diagonal path across a 60-foot by 80-foot park than walking along two edges?

Part 3: Ordering Concrete

Day 3 · Concrete is sold by the cubic yard (1 yd^3 = 27 ft^3).

7. How many cubic yards of concrete would fill the bowl (radius 10 ft, depth 6 ft)? Round to the nearest tenth.

8. How much soil fills 6 cone planters (radius 3 ft, height 4 ft)?

9. Twenty spherical balls of radius 1 foot are made from concrete. How many cubic feet are needed?

Final Product: Skate Park Blueprint

Your final product must include:

- Ramp and rail measurements with converse checks
- Path distances on the map
- Concrete order in cubic yards
- Soil and decoration volumes
- Blueprint with all measurements

Reflection

R1. How does the converse of the Pythagorean Theorem help a builder?

R2. Why must you convert cubic feet to cubic yards by dividing by 27, not 3?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Pythagorean Theorem, distance, and volume calculations are correct and used to make design decisions. Goes beyond the minimum with a creative, realistic choice.	Pythagorean Theorem, distance, and volume calculations are correct and used to make design decisions.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Blueprint and concrete order are labeled with units and show all work. Polished, organized, and visually engaging.	Blueprint and concrete order are labeled with units and show all work. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.